
Lab Assignment # 14

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Task - 1: (Landing Page Design)

• Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

Prompt:

Create a simple landing page for a product or service. Include a header, hero section, and footer. Use HTML and CSS to structure the page.

Code:

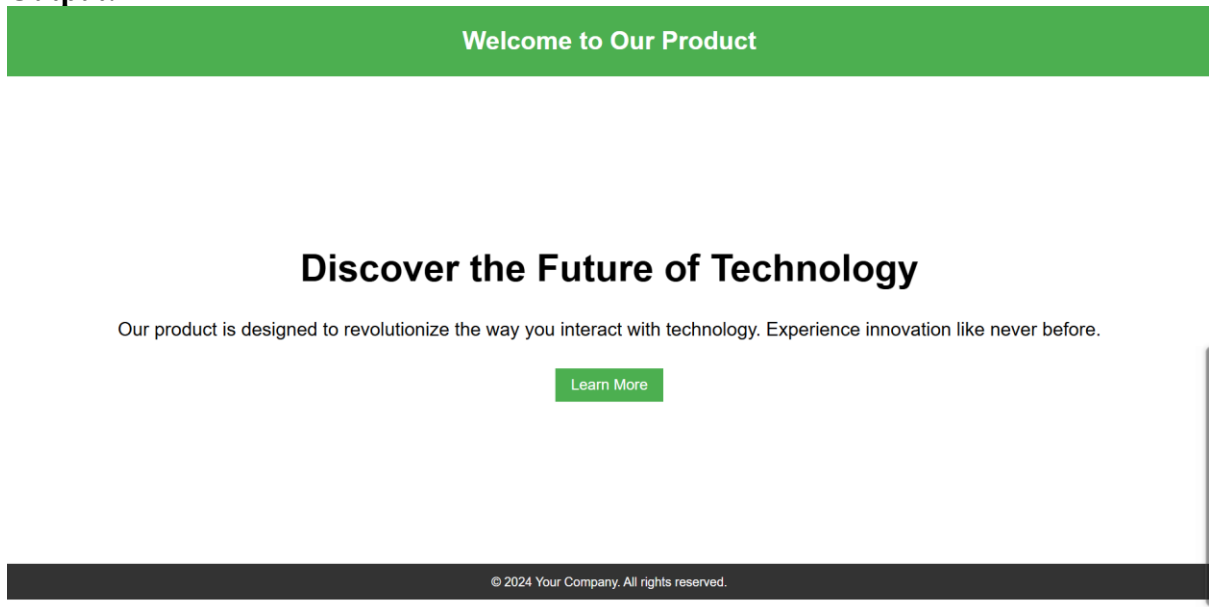
```
Assignment-14.2.html X
AI_AC > Lab > Assignment-14.2.html > ...
1 <!--Create a simple landing page for a product or service. Include a header, hero section, and footer.-->
2 <!--Use HTML and CSS to structure the page.
3 • Use AI to suggest layout alignment and typography.
4 • Ensure the page is responsive across devices.-->
5 <!DOCTYPE html>
6 <html lang="en">
7 <head>
8   <meta charset="UTF-8">
9   <meta name="viewport" content="width=device-width, initial-scale=1.0">
10  <title>Product Landing Page</title>
11  <style>
12    /* Reset some default styles */
13    * {
14      margin: 0;
15      padding: 0;
16      box-sizing: border-box;
17    }
18
19    body {
20      font-family: 'Arial', sans-serif;
21      line-height: 1.6;
22      color: #333;
23    }
24
25    header {
26      background-color: #4CAF50;
27      color: white;
28      padding: 20px 0;
29      text-align: center;
30    }
31
32    .hero {
33      display: flex;
34      flex-direction: column;
35      align-items: center;
36      justify-content: center;
37      height: 80vh;
38      background-image: url('https://via.placeholder.com/1500x800');
```

```

64     footer {
65         background-color: #333;
66         color: white;
67         text-align: center;
68         padding: 10px 0;
69     }
70
71     /* Responsive Design */
72     @media (max-width: 768px) {
73         .hero h1 {
74             font-size: 2em;
75         }
76
77         .hero p {
78             font-size: 1.2em;
79             padding: 0 20px;
80         }
81     }
82 </style>
83 </head>
84 <body>
85     <header>
86         <h1>Welcome to Our Product</h1>
87     </header>
88
89     <section class="hero">
90         <h2>Discover the Future of Technology</h2>
91         <p>Our product is designed to revolutionize the way you interact with technology. Experience innovation like never before.</p>
92         <button>Learn More</button>
93     </section>
94
95     <footer>
96         <p>&copy; 2024 Your Company. All rights reserved.</p>
97     </footer>
98 </body>
99 </html>

```

Output:



Justification:

Showcases a product section with clear headings and bullet points to highlight key features and benefits for users. Focuses on readable layout and persuasive copy to improve user engagement and conversion.

Task – 2: (Theme Toggle Feature)

- Provide a button to switch between light and dark themes.

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

Prompt:

Generate a button that can switch light and dark themes using javascript to toggle css classes and to manage color contrast and accessibility to ensure theme performance instantly on clicking.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Theme Toggle Feature</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      transition: background-color 0.3s, color 0.3s;
    }
    .light-theme {
      background-color: #ffffff;
      color: #000000;
    }
    .dark-theme {
      background-color: #121212;
      color: #ffffff;
    }
    .toggle-button {
      padding: 10px 20px;
      margin: 20px;
      cursor: pointer;
      border: none;
      border-radius: 5px;
      background-color: #0078FF;
      color: #ffffff;
      font-size: 16px;
    }
  </style>
</head>
<body class="light-theme">
  <button class="toggle-button" id="theme-toggle">Toggle Theme</button>

  <script>
    const toggleButton = document.getElementById('theme-toggle');
    const body = document.body;

    toggleButton.addEventListener('click', () => {
      body.classList.toggle('light-theme');
      body.classList.toggle('dark-theme');
    });
  </script>
</body>
</html>
```

Output:

In white Theme

Toggle Theme

In Black Theme

Toggle Theme

Justification:

Implements a theme toggle button that instantly switches between light and dark modes using class-based styling. Improves accessibility and user comfort by adjusting colors dynamically without reloading the page.

Task 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

Prompt:

Generate a user feedback form with fields for Username, Rating, and Comments, ensuring JavaScript validation for required fields and dynamic display of success or error messages.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Feedback Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
    .feedback-form {
      background-color: #fff;
      padding: 20px;
      border-radius: 5px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      width: 300px;
    }
    .feedback-form h2 {
      margin-bottom: 15px;
    }
    .feedback-form input, .feedback-form textarea {
      width: 95%;
      padding: 10px;
      margin-bottom: 10px;
      border: 1px solid #ccc;
      border-radius: 3px;
    }
    .feedback-form button {
      width: 100%;
      padding: 10px;
      background-color: #28a745;
      color: #fff;
      border: none;
      border-radius: 3px;
      cursor: pointer;
    }
    .feedback-form button:hover {
      background-color: #218838;
    }
    .message {
      margin-top: 10px;
    }
```

```
<html lang="en">
<head>
  <style>
  </style>
</head>
<body>
  <div class="feedback-form">
    <h2>User Feedback</h2>
    <input type="text" id="username" placeholder="Username" required>
    <input type="number" id="rating" placeholder="Rating (1-5)" min="1" max="5" required>
    <textarea id="comments" placeholder="Comments" rows="4"></textarea>
    <button onclick="submitFeedback()">Submit</button>
    <div class="message" id="message"></div>
  </div>

  <script>
    function submitFeedback() {
      const username = document.getElementById('username').value.trim();
      const rating = document.getElementById('rating').value.trim();
      const comments = document.getElementById('comments').value.trim();
      const messageDiv = document.getElementById('message');

      // Validate required fields
      if (!username || !rating) {
        messageDiv.textContent = 'Please fill in all required fields.';
        messageDiv.style.color = 'red';
        return;
      }

      // Validate rating value
      if (rating < 1 || rating > 5) {
        messageDiv.textContent = 'Rating must be between 1 and 5.';
        messageDiv.style.color = 'red';
        return;
      }

      // If validation passes, display success message
      messageDiv.textContent = 'Thank you for your feedback!';
      messageDiv.style.color = 'green';

      // Clear the form
      document.getElementById('username').value = '';
      document.getElementById('rating').value = '';
      document.getElementById('comments').value = '';
    }
  </script>
</body>
</html>
```

Output:

User Feedback

User Feedback has been created

Thank you for your feedback!

Justifications:

Builds a feedback form capturing username, rating, and comments with JavaScript validation for required fields. Enhances usability by preventing incomplete submissions and guiding users to correct input errors.

Task - 4 (Image Gallery with JavaScript)

• Task: Display a small image gallery on a web page.

Requirements:

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

Prompt:

Generate a responsive image gallery with next/previous navigation buttons using JavaScript.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Image Gallery</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      display: flex;
      flex-direction: column;
      align-items: center;
      margin: 0;
      padding: 20px;
    }
    .gallery {
      position: relative;
      width: 80%;
      max-width: 600px;
      overflow: hidden;
    }
    .gallery img {
      width: 100%;
      display: none;
    }
    .gallery img.active {
      display: block;
    }
    .navigation {
      margin-top: 10px;
    }
    .navigation button {
      padding: 10px 20px;
      margin: 0 5px;
      cursor: pointer;
    }
    .image-number {
      margin-top: 10px;
    }
    @media (max-width: 600px) {
      .gallery {
        width: 100%;
      }
    }
  </style>
</head>
<body>
  <h1>Responsive Image Gallery</h1>
```

```

<div class="image-number" id="imageNumber">
</div>
<script>
    const images = [
        'https://images.pexels.com/photos/1072179/pexels-photo-1072179.jpeg',
        'https://images.pexels.com/photos/1072180/pexels-photo-1072180.jpeg',
        'https://images.pexels.com/photos/1072181/pexels-photo-1072181.jpeg',
        'https://images.pexels.com/photos/1072182/pexels-photo-1072182.jpeg',
        'https://images.pexels.com/photos/1072183/pexels-photo-1072183.jpeg'
    ];

    const gallery = document.getElementById('gallery');
    const imageNumber = document.getElementById('imageNumber');
    let currentIndex = 0;

    function loadImages() {
        images.forEach((src, index) => {
            const img = document.createElement('img');
            img.src = src;
            if (index === currentIndex) {
                img.classList.add('active');
            }
            gallery.appendChild(img);
        });
    }

    function updateGallery() {
        const imgs = gallery.querySelectorAll('img');
        imgs.forEach((img, index) => {
            img.classList.toggle('active', index === currentIndex);
        });
        imageNumber.textContent = `Image ${currentIndex + 1} of ${images.length}`;
    }

    document.getElementById('prevBtn').addEventListener('click', () => {
        currentIndex = (currentIndex - 1 + images.length) % images.length;
        updateGallery();
    });

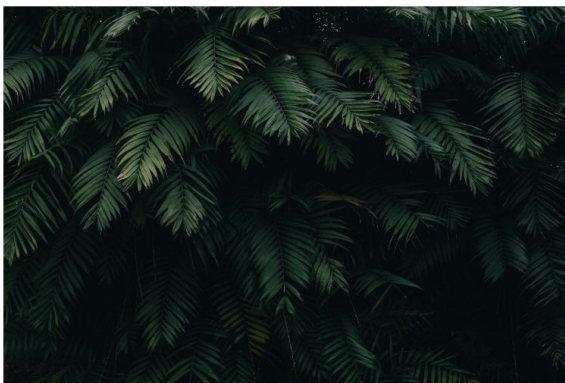
    document.getElementById('nextBtn').addEventListener('click', () => {
        currentIndex = (currentIndex + 1) % images.length;
        updateGallery();
    });

    loadImages();
</script>

```

Output:

Responsive Image Gallery



Previous

Next

Image 1 of 5

Responsive Image Gallery

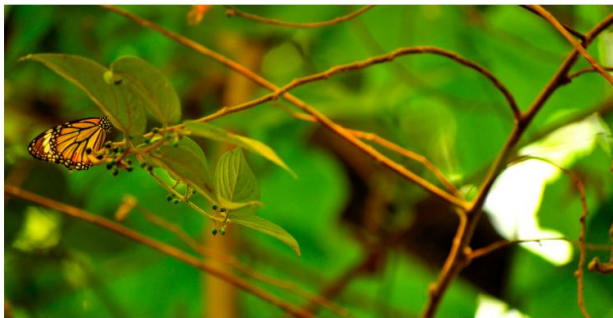


Previous

Next

Image 2 of 5

Responsive Image Gallery



Previous

Next

Image 5 of 5

Justification:

Creates a responsive image gallery that dynamically loads images and supports next/previous navigation buttons. Provides a clean, interactive browsing experience optimized for different screen sizes

Task - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

Prompt:

Create a notification banner that appears at the top of the page and can be dismissed by the user.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Notification Banner</title>
  <style>
    /* Styles for the notification banner */
    #notification-banner {
      position: fixed;
      top: 0;
      left: 0;
      width: 100%;
      background-color: #ffcc00;
      color: #333;
      text-align: center;
      padding: 15px;
      font-size: 18px;
      box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
      display: none; /* Initially hidden */
      z-index: 1000;
    }
    #close-btn {
      margin-left: 20px;
      background-color: #333;
      color: #fff;
      border: none;
      padding: 5px 10px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <div id="notification-banner">
    This is a notification banner!
    <button id="close-btn">Dismiss</button>
  </div>

  <script>
    // Function to show the notification banner
    function showNotification() {
      const banner = document.getElementById('notification-banner');
      banner.style.display = 'block';
      // Add smooth transition
      banner.style.opacity = 0;
      let opacity = 0;
      const fadeIn = setInterval(() => {
```

```
<body>
  <div id="notification-banner">
    This is a notification banner!
    <button id="close-btn">Dismiss</button>
  </div>

  <script>
    // Function to show the notification banner
    function showNotification() {
      const banner = document.getElementById('notification-banner');
      banner.style.display = 'block';
      // Add smooth transition
      banner.style.opacity = 0;
      let opacity = 0;
      const fadeIn = setInterval(() => {
        if (opacity >= 1) {
          clearInterval(fadeIn);
        } else {
          opacity += 0.1;
          banner.style.opacity = opacity;
        }
      }, 50);
    }

    // Function to hide the notification banner
    function hideNotification() {
      const banner = document.getElementById('notification-banner');
      // Add smooth transition
      let opacity = 1;
      const fadeOut = setInterval(() => {
        if (opacity <= 0) {
          clearInterval(fadeOut);
          banner.style.display = 'none';
        } else {
          opacity -= 0.1;
          banner.style.opacity = opacity;
        }
      }, 50);
    }

    // Show the notification banner when the page loads
    window.onload = showNotification;

    // Add event listener to the dismiss button
    document.getElementById('close-btn').addEventListener('click', hideNotification);
  </script>
</body>
</html>
```

Output:

This is a notification banner!

Dismiss

Justification:

Displays a notification banner at the top of the page that can be shown or hidden using JavaScript. Uses smooth transitions and styled layout to deliver important messages without disrupting the overall UI.